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Team ID	SWTID-2026-4593
Team Size	3
Team Leader	Hari Bellamkonda
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Project Name	Machine Learning-Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Project Executable Files

Introduction

The Project Executable Files document describes the executable components, configuration files, trained machine learning models, templates, and supporting resources required to successfully run the Machine Learning-Based Credit Card Approval Prediction System. These files collectively enable application execution, machine learning prediction, user interaction, and database connectivity.

Executable Components

S.No	Executable Component	Description	Purpose
1	app.py	Main Flask application	Starts the web application and coordinates all system modules.
2	model.pkl	Trained Machine Learning Model	Generates credit card approval predictions.
3	encoders.pkl	Label Encoder File	Converts categorical input values into machine-readable numerical values.
4	scaler.pkl	Feature Scaling File	Normalizes input features before prediction.
5	requirements.txt	Python Dependency File	Installs all required project libraries and packages.
6	templates/	HTML Template Folder	Contains the web interface pages for user interaction.
7	static/	CSS, JavaScript and Image Folder	Provides styling, client-side functionality, and static resources.
8	README.md	Project Documentation	Provides installation instructions, project overview, and execution steps.

Execution Requirements

Programming Language	Python 3.x
Web Framework	Flask
Machine Learning Libraries	- Scikit-learn - Pandas - NumPy
Database	MySQL
Development Environment	Visual Studio Code
Version Control	Git and GitHub
Deployment Platform	Render with Docker
Operating System	Windows 11 or Linux

Project Execution Workflow



Key Features of Executable Files

- Modular project structure.
- Easy application deployment.
- Secure machine learning model loading.
- Efficient database integration.
- Responsive web interface.
- Scalable Flask architecture.
- Reusable software components.
- Simplified maintenance and future enhancements.

Conclusion

The executable components of the Machine Learning-Based Credit Card Approval Prediction System provide a complete, modular, and maintainable software solution. Together, the Flask application, trained machine learning model, supporting configuration files, templates, and static resources ensure reliable application execution and accurate real-time credit card approval prediction.